A fixed-two-jet extension lemma for complete strict dg Lie algebras

We use the ordinary cohomological convention. Thus a dg Lie algebra $(\mathfrak{h}, \partial, [-,-])$ has ∂ of degree 1, its bracket has degree 0, and Maurer–Cartan elements have degree 1. No transferred or higher operations occur in this note.

Give

$$s\mathfrak{h}[[s]] = \prod_{j\geq 1} s^j \mathfrak{h}$$

the complete separated filtration $F^q = s^q\mathfrak{h}[[s]]$. Its Maurer–Cartan equation is

$$\partial\alpha + \frac{1}{2}[\alpha, \alpha] = 0.$$

Lemma 1 (fixed-two-jet strict extension)**.** *Let $\mathfrak{h}$ be a dg Lie algebra over a field of characteristic zero. Suppose*

$$\bar{\alpha} = sa_1 + s^2a_2 + s^3a_3 \in \mathrm{MC}(\mathfrak{h} \otimes (s)/(s^4)), \qquad a = a_1.$$

If

$$\ker\big(\mathrm{ad}_a : H^2(\mathfrak{h}) \longrightarrow H^3(\mathfrak{h})\big) = \mathrm{im}\big(\mathrm{ad}_a : H^1(\mathfrak{h}) \longrightarrow H^2(\mathfrak{h})\big), \tag{1}$$

then there is

$$\alpha_\infty \in \mathrm{MC}(s\mathfrak{h}[[s]]) \quad \textit{such that} \quad \alpha_\infty \equiv \bar{\alpha} \pmod{s^3}.$$

In particular, the coefficients of s and s^2 are unchanged.

Proof. For $\gamma \in s\mathfrak{h}^1[[s]]$, put

$$\mathcal{F}(\gamma) = \partial\gamma + \frac{1}{2}[\gamma, \gamma].$$

The dg Lie identities give the strict Bianchi identity

$$\partial\mathcal{F}(\gamma) + [\gamma, \mathcal{F}(\gamma)] = 0. \tag{2}$$

Indeed, since $|\gamma| = 1$,

$$\partial\left(\frac{1}{2}[\gamma, \gamma]\right) = \frac{1}{2}([\partial\gamma, \gamma] - [\gamma, \partial\gamma]) = -[\gamma, \partial\gamma],$$

and graded Jacobi in characteristic zero gives $[\gamma, [\gamma, \gamma]] = 0$.

Choose any lift γ of $\bar{\alpha}$ to $s\mathfrak{h}^1[[s]]$. Its curvature is $O(s^4)$. Suppose inductively that for some $N \geq 4$ we have

$$\mathcal{F}(\gamma) = s^N r_N + s^{N+1} r_{N+1} + O(s^{N+2}), \qquad \gamma = sa + O(s^2). \tag{3}$$

The coefficients of s^N and s^{N+1} in (2) are respectively

$$\partial r_N = 0, \qquad \partial r_{N+1} + [a, r_N] = 0. \tag{4}$$

Thus r_N is a degree-two cocycle and $[a, r_N]$ is a boundary. Equivalently,

$$[r_N] \in \ker\big(\mathrm{ad}_a : H^2(\mathfrak{h}) \longrightarrow H^3(\mathfrak{h})\big).$$

By (1), there is a closed $v \in \mathfrak{h}^1$ for which

$$[r_N] + [a, v] = 0 \quad \text{in } H^2(\mathfrak{h}).$$

Choose $w \in \mathfrak{h}^1$ satisfying

$$r_N + [a, v] = \partial w. \tag{5}$$

Set

$$\delta = s^{N-1}v - s^N w, \qquad \gamma' = \gamma + \delta. \tag{6}$$

The exact curvature-change formula is

$$\mathcal{F}(\gamma + \delta) - \mathcal{F}(\gamma) = \partial\delta + [\gamma, \delta] + \frac{1}{2}[\delta, \delta]. \tag{7}$$

Here $\partial v = 0$, $\gamma - sa = O(s^2)$, and $2N - 2 \geq N + 1$. Consequently the coefficient of s^N on the right of (7) is

$$[a, v] - \partial w.$$

Together with (5), this cancels r_N, while every omitted term has order at least $N + 1$. Hence

$$\mathcal{F}(\gamma') = O(s^{N+1}).$$

Iterating (6) gives a Cauchy sequence. More explicitly, the step indexed by N changes only coefficients in orders at least $N - 1$, so every fixed coefficient eventually stabilizes. Completeness gives a coefficientwise limit α_∞, continuity of ∂ and the bracket gives $\mathcal{F}(\alpha_\infty) \in \bigcap_N s^N\mathfrak{h}[[s]] = 0$, and separatedness is used in the last equality. Since the first correction has order s^3, the limit agrees with $\bar{\alpha}$ modulo s^3. □

Remark 2 (the initial s^4 step)**.** *If the chosen lift has no terms beyond s^3, its first curvature coefficients are*

$$r_4 = [a_1, a_3] + \frac{1}{2}[a_2, a_2], \qquad r_5 = [a_2, a_3].$$

Equation (4) says

$$\partial r_4 = 0, \qquad \partial r_5 + [a_1, r_4] = 0.$$

The correction is $s^3 v_3 - s^4 w_4$. This displays directly why the first and second-order coefficients are fixed.

Remark 3 (what is not preserved)**.** *The conclusion is deliberately congruence modulo s^3, not modulo s^4. At the first step the term $s^3 v_3$ may change a_3. To retain the entire given element modulo s^4, one would need the stronger condition $[r_4] = 0$ in $H^2(\mathfrak{h})$, so that a boundary correction beginning in order s^4 suffices. Exactness (1) alone does not imply this stronger vanishing.*